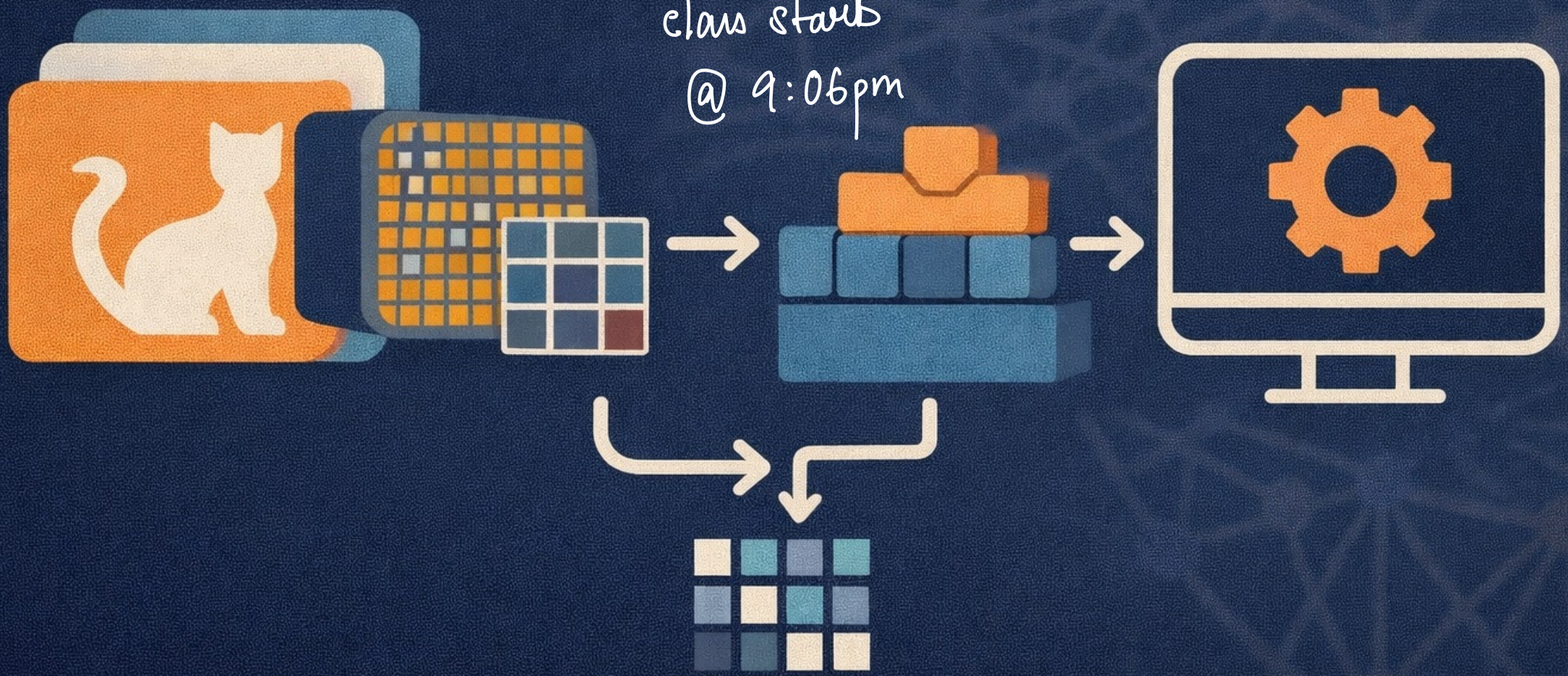
CONVOLUTIONAL NEURAL NETWORK



Lecture : Object detection 3

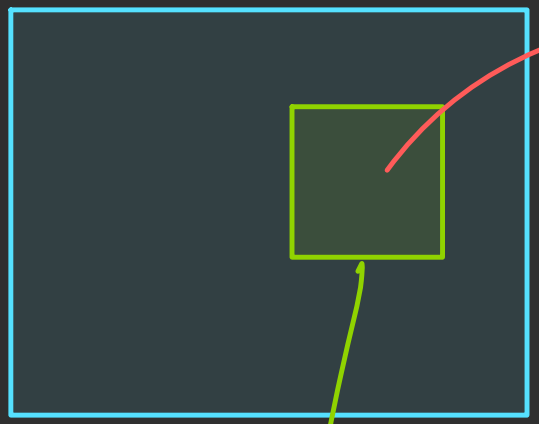
#Agenda

1) Fast RCNN
2) Faster RCNN
3) Single Shot Multi-Box Detector
4) YOLO (You only look Once)

RCNN (Extreamly slow) "Object Detection"

↳ Regional CNN

↳ Inference



→ classification task

→ Regression task

How to solve multiple BB?

↳ NMS

IOU ⇒ Intersection over union

↳ Evalution metric

Where we have to look?

1) sliding window Algo (very slow)

2) Selective Search → ROI

↳ graph based Algo (No training)

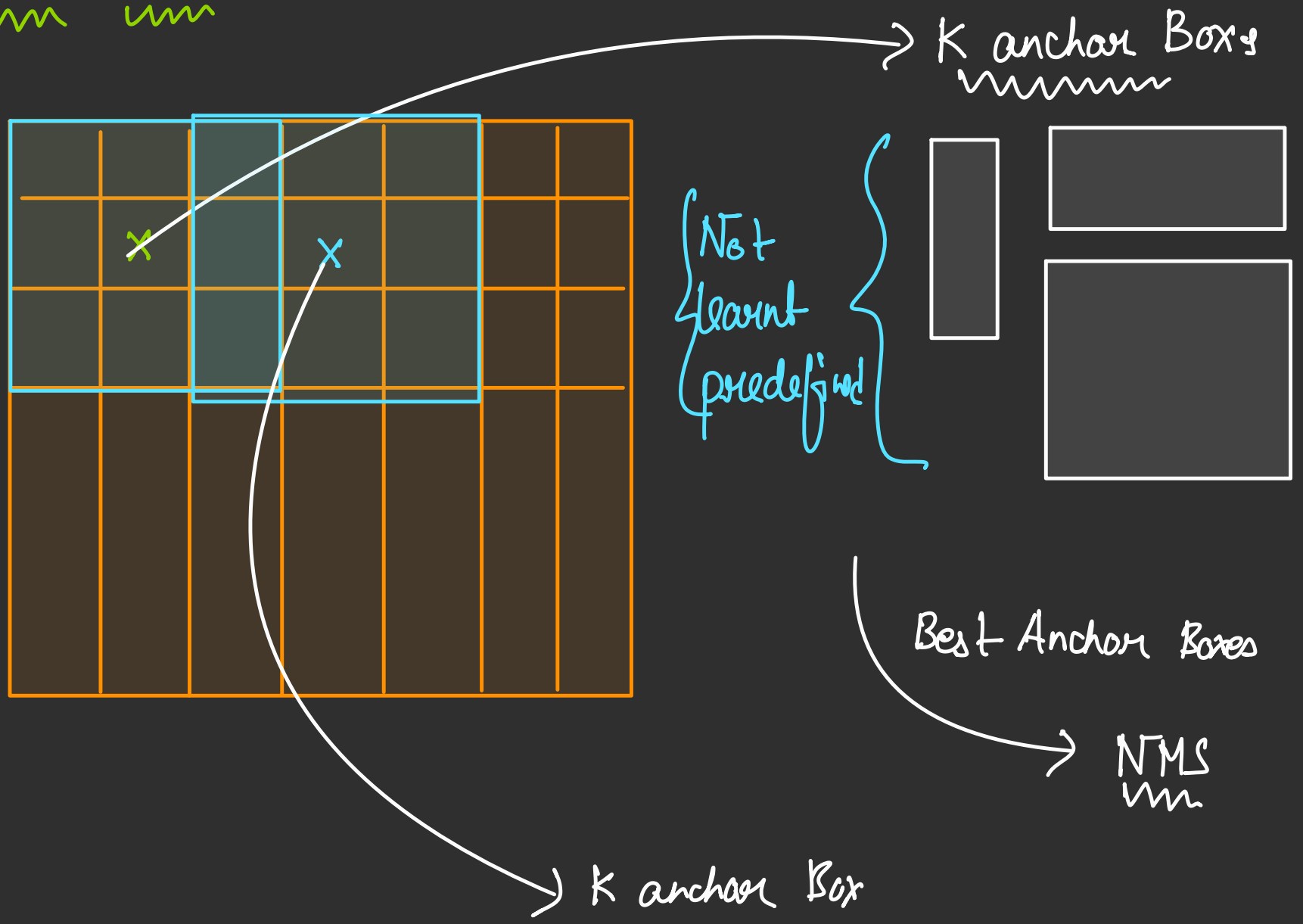
Fast RCNN

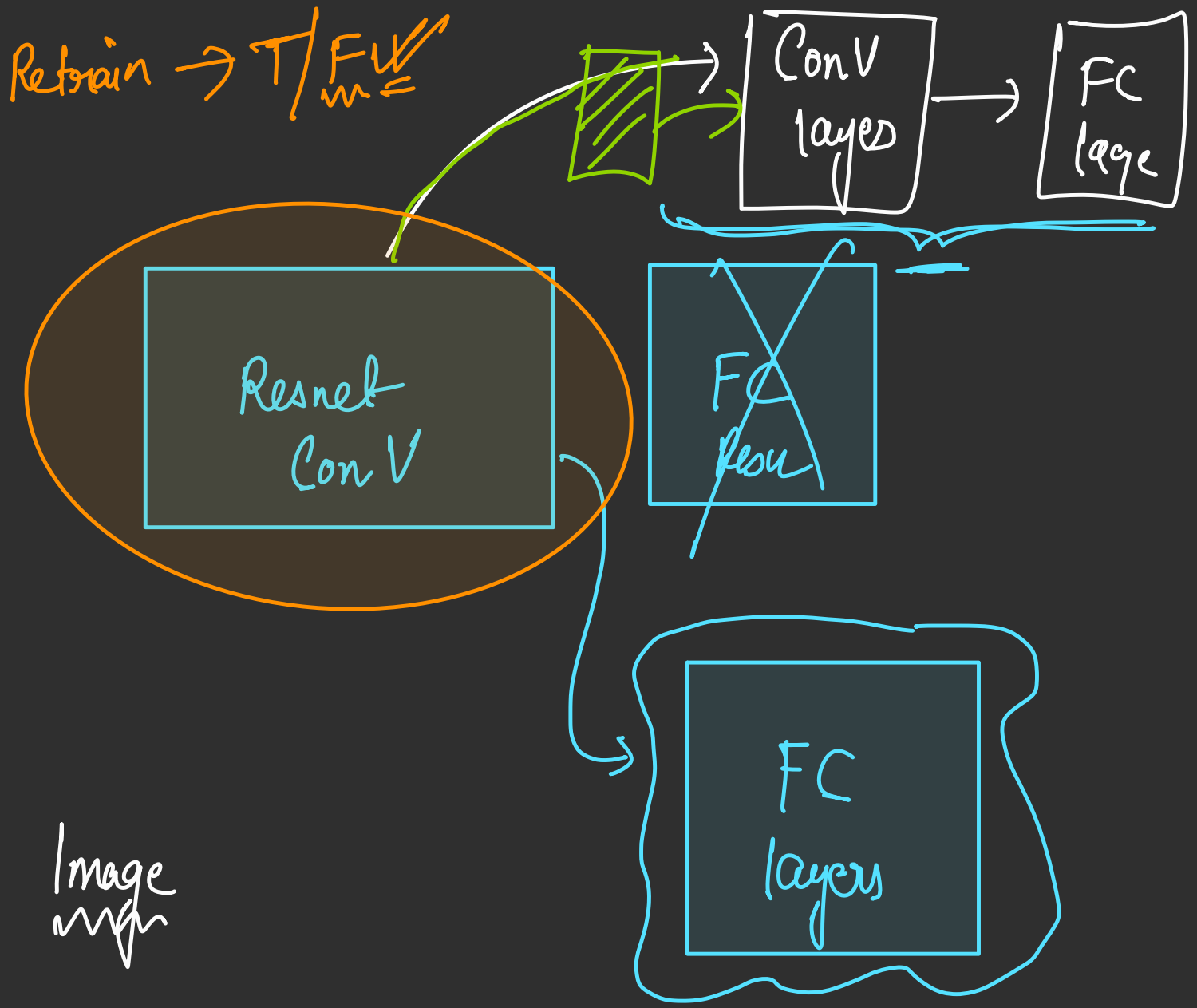
Selective Search → ROI (≈ 2000 proposals)

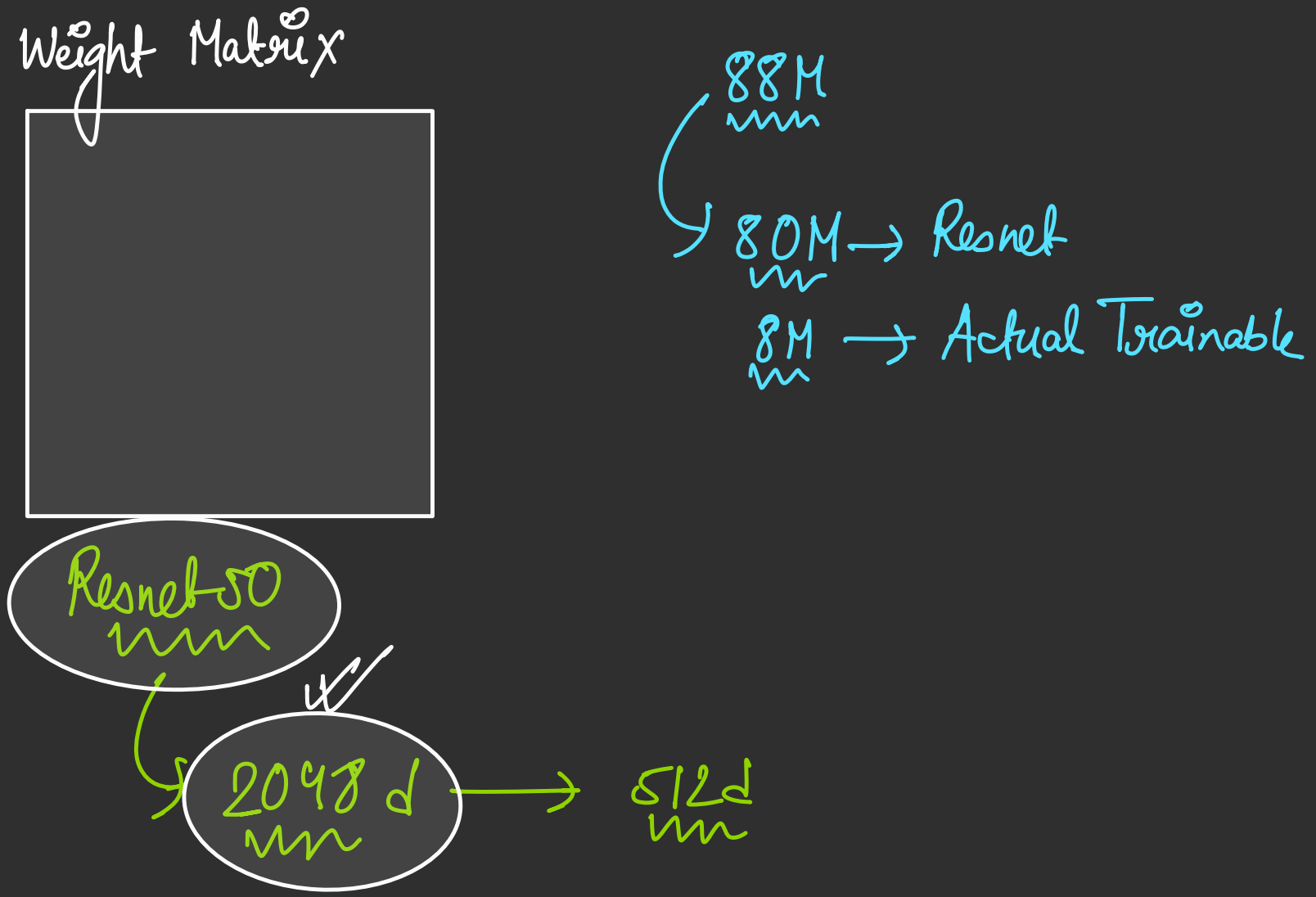
+

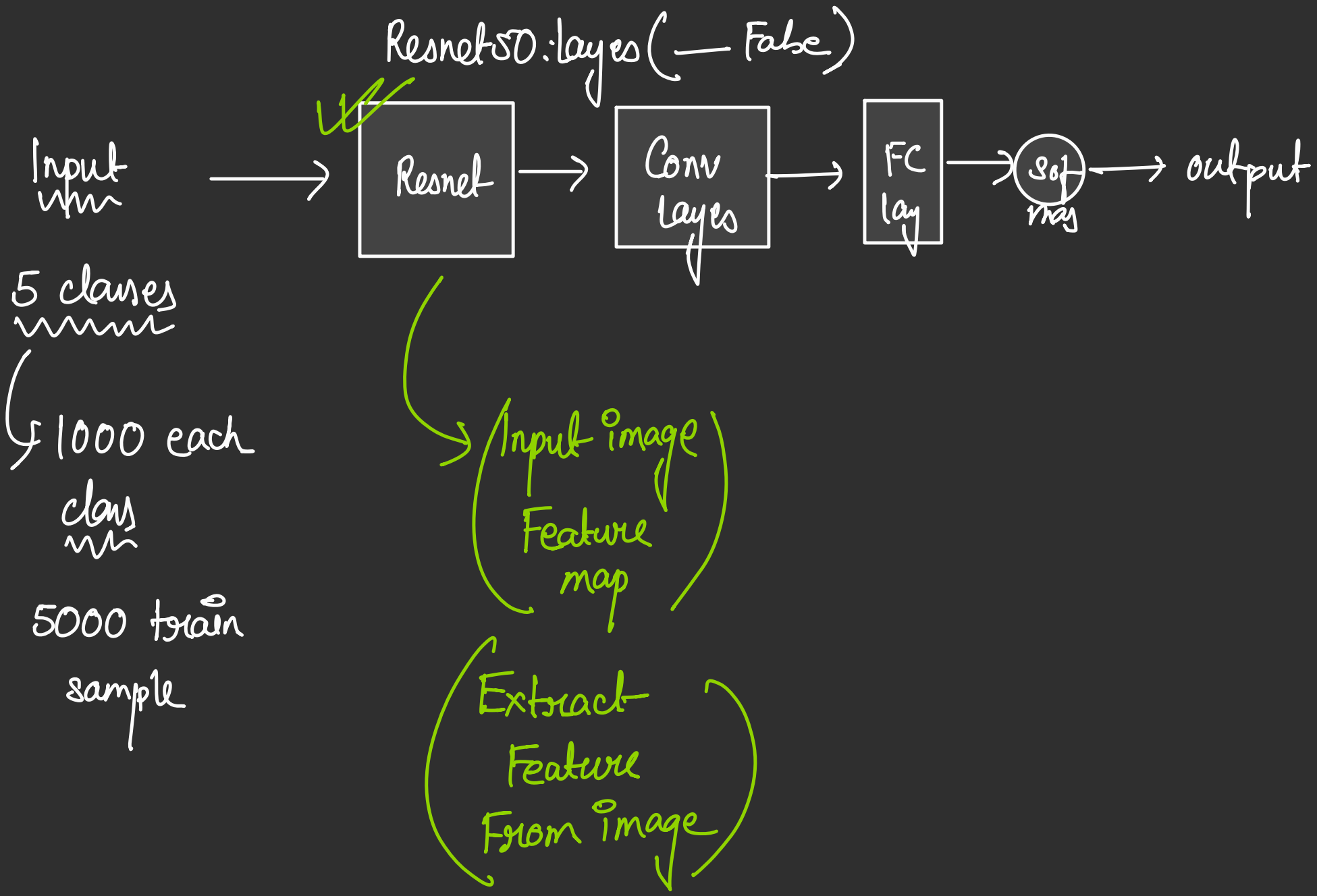
ROI Pooling

Faster RCNN


K anchor Boxes
Not learnt predefined
Best Anchor Boxes
NMS
K anchor Box

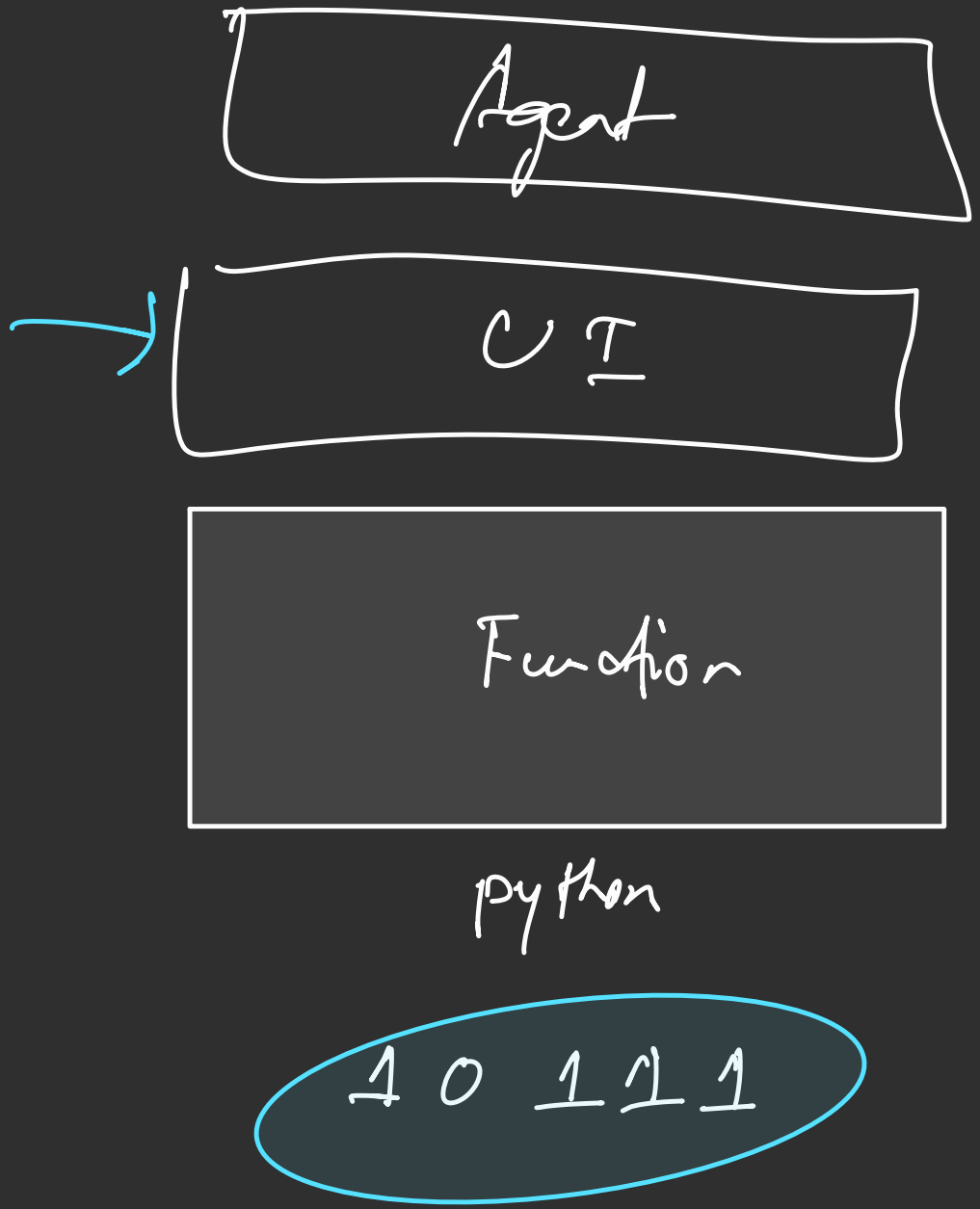

Retrain → T/F
Conv layers
FC layer
Resnet Conv
FC Resn
FC layers
Image


Weight Matrix
Resnet50
2048 d
512 d
88M
80M → Resnet
8M → Actual Trainable


Resnet50.layes(= False)
Input
Resnet
Conv layes
FC lay
Soft max
output
5 clases
1000 each clas
5000 train sample
Input image
Feature map
Extract Feature From image

Key differences

Selective Search Fas


Agent
CLI
Function
python
1 0 1 1 1